Shubhankar Londhe

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Email: shubhankar.londhe@gmail.com, shubhankar.londhe@cit.tum.de **ORCID**: 0000-0002-1220-6650
Website **GitHub** **LinkedIn**

Affiliations

Visiting Researcher 06.2026 - 08.2026
Broad Institute of MIT and Harvard, Cambridge, USA (hosts: Dr. Hilary Finucane, Dr. Konrad Karczewski)

Doctoral Researcher 01.07.2024 - Present
Helmholtz Munich

Doctoral Researcher 01.10.2022 - Present
Technical University of Munich

Education

PhD in Computational Biology October 2022 - Present
Technical University of Munich & Helmholtz Munich

- **Supervisor: Prof. Dr. Julien Gagneur**
- **Research Focus:** Developing methods to capture rare variant effects in diseases and complex traits. Working with biobank-scale data (UK Biobank, All of Us, Genomics England) and large rare-disease cohorts (Solve-RD).
- Enrolled in the prestigious **Munich School for Data Science (MUDS)** PhD program
- Supervised 3 Master's thesis, and multiple graduate and undergraduate student interns.

Master of Science August 2020 - May 2022
Indian Institute of Science Education and Research, Pune

- Majored in Mathematics, GPA: 8.5 (out of 10)
- **Master thesis** (Supervisor: Dr. Annalisa Marsico)
  Implemented explainable AI methods to quantify secondary structure binding preferences of RNA-Binding Protein

Bachelor of Science August 2017 - May 2020
Indian Institute of Science Education and Research, Pune

- Majored in Mathematics, GPA: 8.5 (out of 10)
- **Research assistant** (Supervisor: Prof. M. S. Madhusudhan)
  Developed an algorithm to learn structural patterns from protein micro-environments using deep learning.

Publications

1. **Shubhankar Londhe**, Eva Holtkamp, Georgios Tsitsiridis, Anna Starovoit, Pedro Tomaz da Silva, Johannes Hingerl, Hilary Finucane, Julien Gagneur. "Phenotypes of ultra-rare variant carriers benchmark variant effect scores". ***bioRxiv*** (2026).
2. Nils Wagner, Aleksandr M Neverov, Alexandra C Martin-Geary, **Shubhankar Londhe**, Carina Schröder, Vicente A Yépez, Nicola Whiffin, Julien Gagneur. "Aberrant splicing prediction during human organ development". ***bioRxiv*** (2025).
3. Vicente A Yépez, German Demidov, Kornelia Ellwanger, ... **Shubhankar Londhe**, ... Solve-RD consortium, Julien Gagneur, Holm Graessner. "The Solve-RD Solvathons as a pan-European interdisciplinary collaboration to diagnose patients with rare disease". ***Nature Genetics*** (2025).
4. **Shubhankar Londhe**, Jonas Lindner, Zhifen Chen, Eva Holtkamp, Florian R. Holzlwimmer, Francesco Paolo Casale, Felix Brechtmann, and Julien Gagneur. "Functional gene embeddings improve rare variant polygenic risk scores". ***bioRxiv*** (2024).

5. Felix Brechtmann, Thibault Bechtler, **Shubhankar Londhe**, Christian Mertes, and Julien Gagneur. "Evaluation of input data modality choices on functional gene embeddings". ***NAR Genomics and Bioinformatics*** (2023)
6. Marc Horlacher, Giulia Cantini, Julian Hesse, Patrick Schinke, Nicolas Goedert, **Shubhankar Londhe**, Lambert Moyon, and Annalisa Marsico. "A systematic benchmark of machine learning methods for protein–RNA interaction prediction". ***Briefings in Bioinformatics*** (2023)

Talks

1. **ERDERA Science Club Series** October 2025
   Functional gene embeddings improve rare variant based gene discovery
2. **Annual meeting of the European Society of Human Genetics (ESHG), Milan, Italy** May 2025
   Functional gene embeddings improve rare variant polygenic risk scores
3. **Weihenstephan Bioinformatics Symposium, Freising, Germany** April 2025
   Functional gene embeddings improve rare variant polygenic risk scores
4. **Annual meeting of the American Society of Human Genetics (ASHG), Denver, USA** November 2024
   Functional gene embeddings improve rare variant polygenic risk scores

Skills

- **Programming**
  5+ years Python (PyTorch, pandas, optuna, etc.), LaTeX
  3+ years GPU computing, Snakemake, Git, SLURM, bash, R (ggplot, tidyverse, etc.)
- **Data & Machine Learning**
  Patient and biobank-scale genetic data (UK Biobank, All of Us, Genomics England, Solve-RD)
  Variant call files (VCF) and biobank cloud infrastructure (e.g. UK Biobank RAP / DNAnexus)
  Training and evaluating machine learning models (PyTorch, GPU computing)
- **Teaching** (teaching team, *TUM*)

Machine Learning for Human Genetics	October 2026 - Present
A Practical Introduction to Data Science	October 2026 - Present
Computational Modelling for System Genetics	October 2022 - April 2026
Data Analysis and Visualization in R	October 2022 - April 2026
Machine Learning for Regulatory Genomics	April 2023 - October 2025

- **Languages**
  Native: English, Marathi | **Fluent:** Hindi

Grants & Achievements

- **EMBO Scientific Exchange Grant #12502** June - August 2026
  3-month research exchange at the Broad Institute of MIT and Harvard with Hilary Finucane and Konrad Karczewski.
- **iGEM (International Genetically Engineered Machine), Silver Medal** November 2019
  Performed the quantitative modelling of the engineered synthetic operon system in the bacteria.
- **All India Rank 51** in the all-India level IISER Aptitude Test July 2017

References

Available on request.